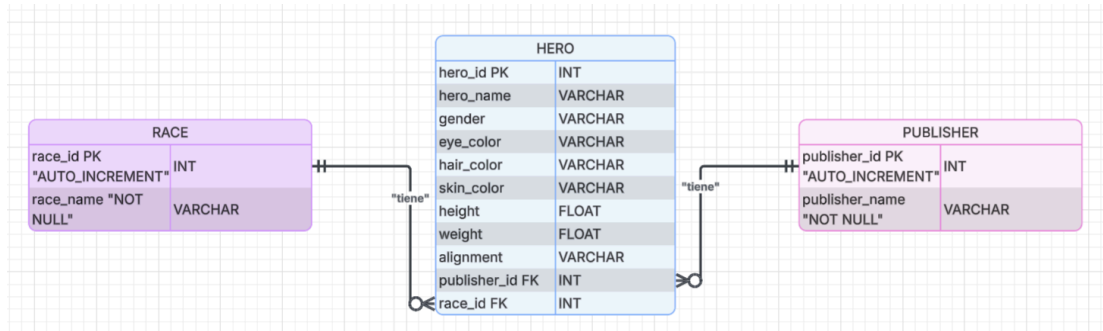


RETO SQL

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Manipulación de Bases de Datos SQL

2. Modelo relacional



3. Crea la estructura de la base de datos

```
CREATE DATABASE heroes;
```

```
USE heroes;
```

```
CREATE TABLE publisher(  
    publisher_id INT NOT NULL AUTO_INCREMENT PRIMARY KEY,  
    publisher_name VARCHAR(100) NOT NULL  
);  
DESCRIBE publisher;
```

```
CREATE TABLE race(  
    race_id INT NOT NULL AUTO_INCREMENT PRIMARY KEY,  
    race_name VARCHAR(100) NOT NULL  
);  
DESCRIBE race;
```

```
CREATE TABLE hero (  
    hero_id INTEGER PRIMARY KEY,  
    hero_name VARCHAR(100),  
    gender VARCHAR(20),  
    eye_color VARCHAR(50),  
    hair_color VARCHAR(50),  
    skin_color VARCHAR(50),  
    height FLOAT,  
    weight FLOAT,
```

```

alignment VARCHAR(20),

publisher_id INTEGER,
race_id INTEGER,

FOREIGN KEY (publisher_id) REFERENCES publisher(publisher_id),

FOREIGN KEY (race_id) REFERENCES race(race_id)
);
DESCRIBE hero;

```

Creando base de datos, tablas y describe de cada una:

1	20:07:46	DROP DATABASE IF EXISTS heroes	3 row(s) affected
2	20:07:52	CREATE DATABASE heroes	1 row(s) affected
3	20:07:58	USE heroes	0 row(s) affected
4	20:08:58	CREATE TABLE publisher(publisher_id INT NOT NULL AUTO_INCREMENT PRIMARY KEY, publisher_name VARCHAR(100) NOT NULL)	0 row(s) affected
5	20:09:02	CREATE TABLE race(race_id INT NOT NULL AUTO_INCREMENT PRIMARY KEY, race_name VARCHAR(100) NOT NULL)	0 row(s) affected
6	20:09:06	CREATE TABLE hero(hero_id INT NOT NULL AUTO_INCREMENT PRIMARY KEY, hero_name VARCHAR(100), gender enum('Male','Female'), eye_color VARCHAR(50), hair_color VARCHAR(50), height FLO...	0 row(s) affected
10	20:11:59	DESCRIBE publisher	2 row(s) returned
11	20:12:04	DESCRIBE race	2 row(s) returned
12	20:12:26	DESCRIBE hero	11 row(s) returned

Tabla publisher:

Field	Type	Null	Key	Default	Extra
publisher_id	int(11)	NO	PRI	NULL	auto_increment
publisher_name	varchar(100)	NO		NULL	

Tabla race:

Field	Type	Null	Key	Default	Extra
race_id	int(11)	NO	PRI	NULL	auto_increment
race_name	varchar(100)	NO		NULL	

Tabla hero:

Field	Type	Null	Key	Default	Extra
hero_id	int(11)	NO	PRI	NULL	auto_increment
hero_name	varchar(100)	YES		NULL	
gender	enum('Male','Female')	YES		NULL	
eye_color	varchar(50)	YES		NULL	
hair_color	varchar(50)	YES		NULL	
height	float	YES		NULL	
skin_color	varchar(50)	YES		NULL	
alignment	varchar(20)	YES		NULL	
weight	float	YES		NULL	
publisher_id	int(11)	YES	MUL	NULL	
race_id	int(11)	YES	MUL	NULL	

4. Inserta los primeros 20 registros

```

✓ 31 20:25:35 INSERT INTO publisher VALUES (1,'Marvel Comics'), (2,'Dark... 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
✓ 32 20:25:38 INSERT INTO race VALUES (1,'Human'), (2,'Ichthy Sapien'), (... 9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
✓ 33 20:25:41 INSERT INTO hero VALUES (1,'A-Bomb','Male','yellow','No H... 20 row(s) affected Records: 20 Duplicates: 0 Warnings: 0

```

```

INSERT INTO publisher VALUES (1,'Marvel Comics'),
(2,'Dark Horse Comics'),
(3,'DC Comics'),
(4,'NBC - Heroes'),
(5,'Wildstorm');

```

```

INSERT INTO race VALUES (1,'Human'),
(2,'Ichthy Sapien'),
(3,'Ungaran'),
(4,'Human / Radiation'),
(5,'Cosmic Entity'),
(6,'-'),
(7,'Cyborg'),
(8,'Xenomorph XX121'),
(9,'Android');

```

```

INSERT INTO hero VALUES
(1,'A-Bomb','Male','yellow','No Hair','-','203,441','good',1,1),
(2,'Abe Sapien','Male','blue','No Hair','blue',191,65,'good',2,2),
(3,'Abin Sur','Male','blue','No Hair','red',185,90,'good',3,3),
(4,'Abomination','Male','green','No Hair','-','203,441','bad',1,4),
(5,'Abraxas','Male','blue','Black','-','-99,NULL,'bad',1,5),
(6,'Absorbing Man','Male','blue','No Hair','-','193,122','bad',1,1),
(7,'Adam Monroe','Male','blue','Blond','-','-99,NULL,'good',4,6),
(8,'Adam Strange','Male','blue','Blond','-','185,88','good',3,1),
(9,'Agent 13','Female','blue','Blond','-','173,61','good',1,6),
(10,'Agent Bob','Male','brown','Brown','-','178,81','good',1,1),
(11,'Agent Zero','Male','-','-','-','191,104','good',1,6),
(12,'Air-Walker','Male','blue','White','-','188,108','bad',1,6),
(13,'Ajax','Male','brown','Black','-','193,90','bad',1,7),
(14,'Alan Scott','Male','blue','Blond','-','180,90','good',3,6),
(15,'Alex Mercer','Male','-','-','-','-99,NULL,'bad',5,1),
(16,'Alex Woolsey','Male','-','-','-','-99,NULL,'good',4,6),
(17,'Alfred Pennyworth','Male','blue','Black','-','178,72','good',3,1),
(18,'Alien','Male','-','No Hair','black',244,169,'bad',2,8),
(19,'Allan Quatermain','Male','-','-','-','-99,NULL,'good',5,6),
(20,'Amazo','Male','red','-','-','257,173','bad',3,9);

```

5. Modifica los valores incorrectos de la columna Weight

```

UPDATE hero SET weight = NULL
WHERE weight = -99.0;

```

SELECT weight FROM hero;

	weight
▶	441
	65
	90
	441
	NULL
	122
	NULL
	88
	61
	81
	104
	108
	90
	90
	NULL
	NULL
	72
	169
	NULL
	173

✓	42	20:49:38	UPDATE hero SET weight = NULL WHERE weight = -99.0	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0
✓	43	20:50:04	SELECT weight FROM hero LIMIT 0, 1000	20 row(s) returned

6. Recrea la información original con un JOIN

Consulta:

```
SELECT
  h.hero_id,
  h.hero_name,
  h.gender,
  h.eye_color,
  r.race_name,
  h.hair_color,
  h.height,
  p.publisher_name,
  h.skin_color,
  h.alignment,
  h.weight
FROM hero h
JOIN publisher p
  ON h.publisher_id = p.publisher_id
JOIN race r
  ON h.race_id = r.race_id
ORDER BY h.hero_id;
```

	hero_id	hero_name	gender	eye_color	race_name	hair_color	height	publisher_name	skin_color	alignment	weight
▶	1	A-Bomb	Male	yellow	Human	No Hair	203	Marvel Comics	-	good	441
	2	Abe Sapien	Male	blue	Ichthyo Sapien	No Hair	191	Dark Horse Comics	blue	good	65
	3	Abin Sur	Male	blue	Ungaran	No Hair	185	DC Comics	red	good	90
	4	Abomination	Male	green	Human / Radiation	No Hair	203	Marvel Comics	-	bad	441
	5	Abraxas	Male	blue	Cosmic Entity	Black	-99	Marvel Comics	-	bad	NULL
	6	Absorbing Man	Male	blue	Human	No Hair	193	Marvel Comics	-	bad	122
	7	Adam Monroe	Male	blue	-	Blond	-99	NBC - Heroes	-	good	NULL
	8	Adam Strange	Male	blue	Human	Blond	185	DC Comics	-	good	88
	9	Agent 13	Female	blue	-	Blond	173	Marvel Comics	-	good	61
	10	Agent Bob	Male	brown	Human	Brown	178	Marvel Comics	-	good	81
	11	Agent Zero	Male	-	-	-	191	Marvel Comics	-	good	104
	12	Air-Walker	Male	blue	-	White	188	Marvel Comics	-	bad	108
	13	Ajax	Male	brown	Cyborg	Black	193	Marvel Comics	-	bad	90
	14	Alan Scott	Male	blue	-	Blond	180	DC Comics	-	good	90
	15	Alex Mercer	Male	-	Human	-	-99	Wildstorm	-	bad	NULL
	16	Alex Woosly	Male	-	-	-	-99	NBC - Heroes	-	good	NULL
	17	Alfred Penny...	Male	blue	Human	Black	178	DC Comics	-	good	72
	18	Alien	Male	-	Xenomorph XX121	No Hair	244	Dark Horse Comics	black	bad	169
	19	Allan Quaterm...	Male	-	-	-	-99	Wildstorm	-	good	NULL
	20	Amazo	Male	red	Android	-	257	DC Comics	-	bad	173

✓ 48 20:56:51 SELECT h.hero_id, h.hero_name, h.gender, h.eye_... 20 row(s) returned

Vista:

CREATE VIEW superhero AS

SELECT

h.hero_id,
h.hero_name,
h.gender,
h.eye_color,
r.race_name,
h.hair_color,
h.height,
p.publisher_name,
h.skin_color,
h.alignment,
h.weight

FROM hero h

JOIN publisher p

ON h.publisher_id = p.publisher_id

JOIN race r

ON h.race_id = r.race_id

ORDER BY h.hero_id;

	hero_id	hero_name	gender	eye_color	race_name	hair_color	height	publisher_name	skin_color	alignment	weight
▶	1	A-Bomb	Male	yellow	Human	No Hair	203	Marvel Comics	-	good	441
	2	Abe Sapien	Male	blue	Ichthy Sapien	No Hair	191	Dark Horse Comics	blue	good	65
	3	Abin Sur	Male	blue	Ungaran	No Hair	185	DC Comics	red	good	90
	4	Abomination	Male	green	Human / Radiation	No Hair	203	Marvel Comics	-	bad	441
	5	Abraxas	Male	blue	Cosmic Entity	Black	-99	Marvel Comics	-	bad	NULL
	6	Absorbing Man	Male	blue	Human	No Hair	193	Marvel Comics	-	bad	122
	7	Adam Monroe	Male	blue	-	Blond	-99	NBC - Heroes	-	good	NULL
	8	Adam Strange	Male	blue	Human	Blond	185	DC Comics	-	good	88
	9	Agent 13	Female	blue	-	Blond	173	Marvel Comics	-	good	61
	10	Agent Bob	Male	brown	Human	Brown	178	Marvel Comics	-	good	81
	11	Agent Zero	Male	-	-	-	191	Marvel Comics	-	good	104
	12	Air-Walker	Male	blue	-	White	188	Marvel Comics	-	bad	108
	13	Ajax	Male	brown	Cyborg	Black	193	Marvel Comics	-	bad	90
	14	Alan Scott	Male	blue	-	Blond	180	DC Comics	-	good	90
	15	Alex Mercer	Male	-	Human	-	-99	Wildstorm	-	bad	NULL
	16	Alex Woosly	Male	-	-	-	-99	NBC - Heroes	-	good	NULL
	17	Alfred Penny...	Male	blue	Human	Black	178	DC Comics	-	good	72
	18	Alien	Male	-	Xenomorph XX121	No Hair	244	Dark Horse Comics	black	bad	169
	19	Allan Quaterm...	Male	-	-	-	-99	Wildstorm	-	good	NULL
	20	Amazo	Male	red	Android	-	257	DC Comics	-	bad	173

✓ 49 20:58:31 CREATE VIEW superhero AS SELECT h.hero_id, h.hero... 0 row(s) affected

✓ 50 20:58:41 SELECT * FROM superhero LIMIT 0, 1000 20 row(s) returned

7. Consulta al superhéroe más alto

SELECT *, MAX(height) FROM hero;

	hero_id	hero_name	gender	eye_color	hair_color	skin_color	height	weight	alignment	publisher_id	race_id	MAX(height)
▶	1	A-Bomb	Male	yellow	No Hair	-	203	441	good	1	1	257

✓ 52 21:00:18 SELECT *, MAX(height) FROM hero LIMIT 0, 1000 1 row(s) returned

8. Consulta a los superhéroes con peso mayor al promedio

```
SELECT * FROM hero
WHERE weight > (
    SELECT AVG(weight) FROM hero
);
```

	hero_id	hero_name	gender	eye_color	hair_color	skin_color	height	weight	alignment	publisher_id	race_id
▶	1	A-Bomb	Male	yellow	No Hair	-	203	441	good	1	1
	4	Abomination	Male	green	No Hair	-	203	441	bad	1	4
	18	Alien	Male	-	No Hair	black	244	169	bad	2	8
	20	Amazo	Male	red	-	-	257	173	bad	3	9
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

✓ 54 21:03:54 SELECT * FROM hero WHERE weight > (SELECT AVG(weig... 4 row(s) returned

9. Consulta el promedio de estatura por editorial

```
SELECT publisher_name, AVG(height) FROM superhero
GROUP BY publisher_name;
```

	publisher_name	AVG(height)
▶	Marvel Comics	158.11111111111111
	Dark Horse Comics	217.5
	DC Comics	197
	NBC - Heroes	-99
	Wildstorm	-99

✓ 56 21:07:09 SELECT publisher_name, AVG(height) FROM superhero GRO... 5 row(s) returned

10. Reproduce la consulta anterior en Pandas usando pandaSQL

```
[2]
✓ 10 s !pip install pandasql

> ... Mostrar el resultado oculto

[3]
✓ 1 s import pandas as pd
from pandasql import sqldf

[5]
✓ 0 s df = pd.read_csv('heroes_information.csv')

query = """
    SELECT Publisher, AVG(Height) FROM df
    GROUP BY Publisher;
    """

resultado = sqldf(query)

print(resultado)
```

	Publisher	AVG(Height)
0	None	-2.366667
1	ABC Studios	-99.000000
2	DC Comics	91.072093
3	Dark Horse Comics	69.611111
4	George Lucas	143.500000
5	Hanna-Barbera	-99.000000
6	HarperCollins	-99.000000
7	IDW Publishing	-99.000000
8	Icon Comics	-99.000000
9	Image Comics	-76.857143
10	J. K. Rowling	-99.000000
11	J. R. R. Tolkien	279.000000
12	Marvel Comics	142.756443
13	Microsoft	213.000000
14	NBC - Heroes	-99.000000
15	Rebellion	188.000000
16	Shueisha	171.500000
17	Sony Pictures	179.000000
18	South Park	-99.000000
19	Star Trek	-5.500000
20	SyFy	-99.000000
21	Team Epic TV	182.200000
22	Titan Books	183.000000
23	Universal Studios	-99.000000
24	Wildstorm	-99.000000

https://colab.research.google.com/drive/1T2ZsQUZhXqzSShXKmWxry0zba1h6s0Y3?usp=s_haring